

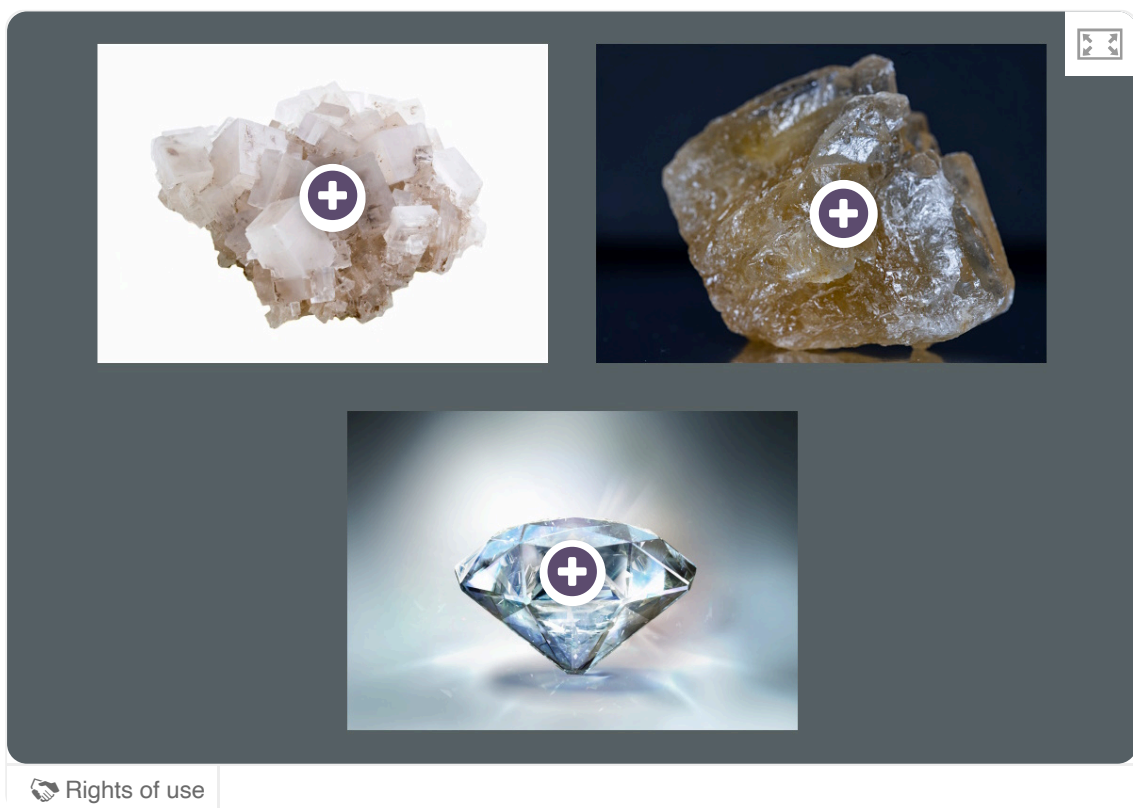
Metallic bonding and structure

Learning outcomes

By the end of this section you should be able to:

- Define metallic bonding.
- Describe the structure of the metallic lattice.
- Sketch a diagram showing the lattice structure of a metal.

When we think of crystalline solids, we typically think of ionic and covalent solids (e.g. sodium chloride, sugar and diamond). A crystalline solid is where the constituents are arranged in a highly ordered lattice structure that extends in all directions. **Interactive 1** shows some of these examples of ionic and covalent crystals.



Interactive 1. Crystalline Solids of Sodium Chloride, Sugar and Diamond.

 More information for interactive 1

An interactive collage displays three photographs of crystalline solids. Each photograph has a plus sign representing an interactive hotspot. Selecting a hotspot reveals the internal crystal structure of the corresponding material.

Top left:

Photo: A chunk of rock salt displaying intergrown white to transparent crystals.

Hotspot: A cubic structure of sodium chloride featuring an alternating arrangement of sodium and chloride ions. Chloride ions are represented by large spheres, and sodium ions are represented by small spheres.

Top right:

Photo: An irregularly shaped golden brown sugar crystal.

Hotspot: The molecular structure of sugar depicts repeating sucrose units with carbon, hydrogen, and oxygen atoms.

Bottom:

Photo: A single crystal of diamond displaying its shiny, reflective surface.

Hotspot: A cubic structure illustrates carbon atoms occupying the corners and face centers of the cube. Each carbon atom is bonded to four other carbon atoms in a tetrahedral geometry.

This interactive collage helps users identify and recognize the external appearance and internal structure of three crystalline substances.

We have seen examples of crystalline substances that have ionic and covalent bonding. Did you know that metals also exist as crystalline solids? **Figure 1** shows an example of bismuth metal that crystallises in a unique staircase structure. As you learn more about metals, consider the structure and bonding of metals compared to ionic salts and covalent network crystals.

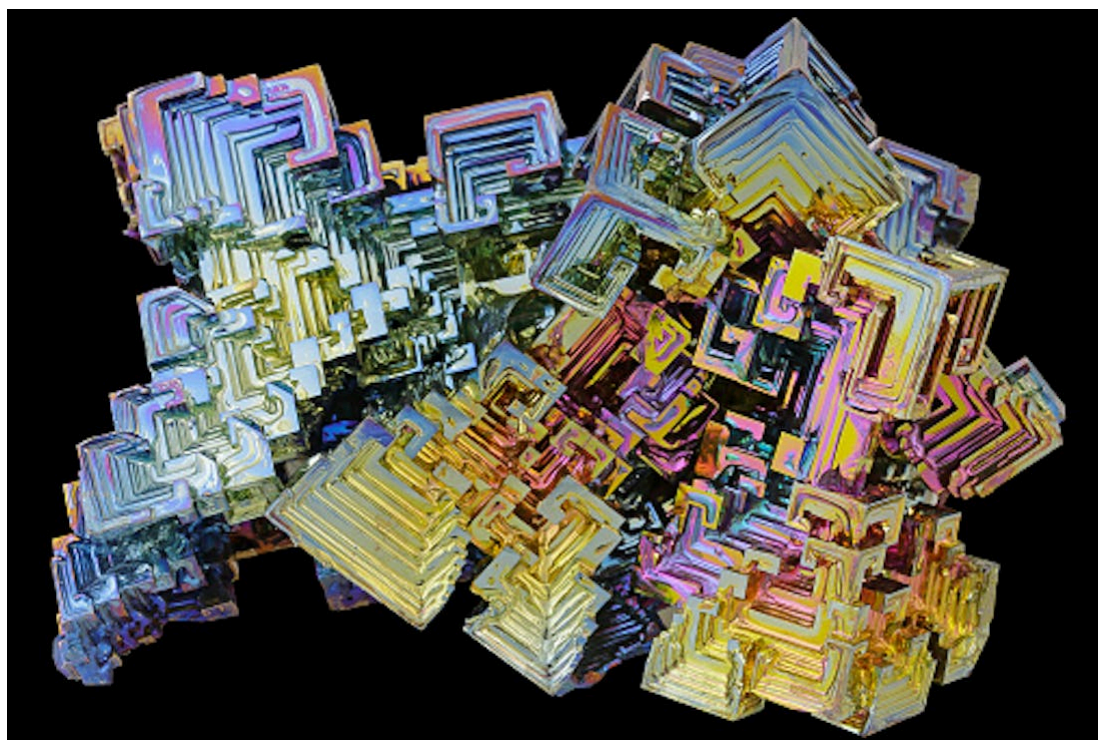


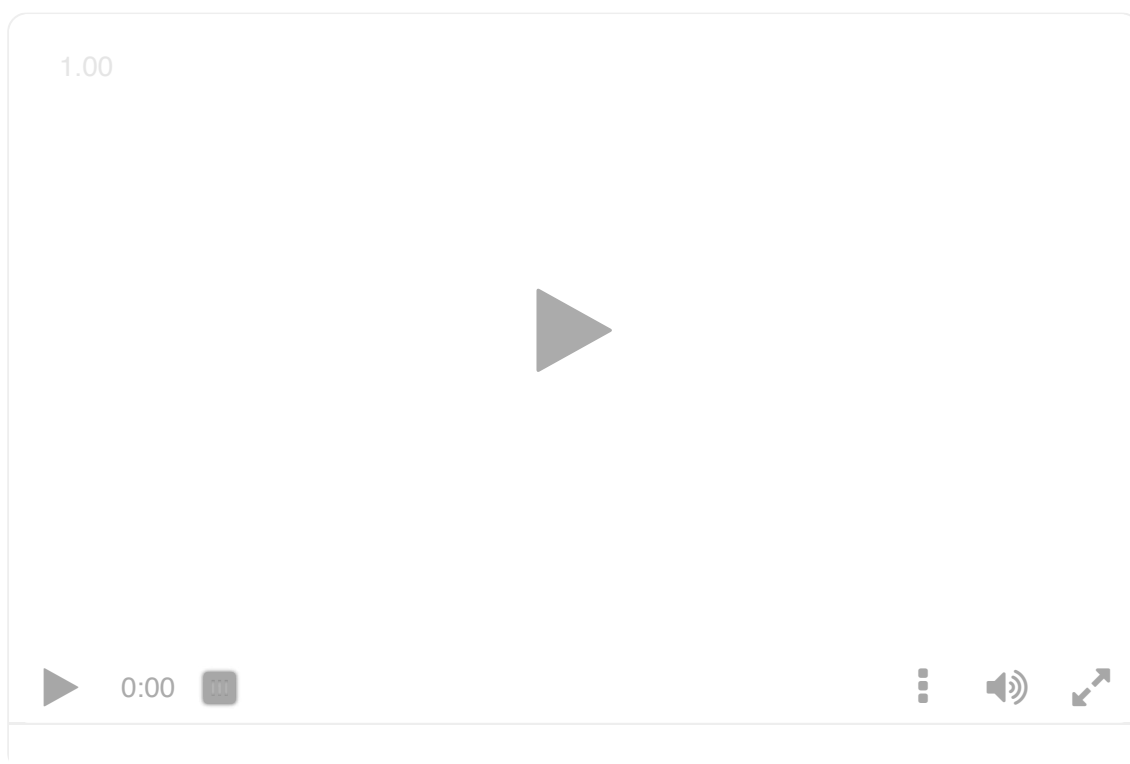
Figure 1. Bismuth crystals.

Credit: Walter Geiersperger, Getty Images

(<https://www.gettyimages.co.uk/license/589145472>)

In section S2.1.2 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ionic-bonding-id-43557/>) you learned how ionic bonding resulted from the electrostatic attraction between oppositely charged ions. Metals are neutral atoms. How are uncharged atoms able to attract one another to form a bond? What could be occurring within the metallic bond to allow neutral particles to remain attracted together as a solid?

Let us look at the formation of the metallic bond by considering the metallic element lithium in **Interactive 2**.



Interactive 2. Formation of the Metallic Bond for Lithium.

 More information for interactive 2

A video displays four rows and four columns of lithium atoms. The text on the left reads as follows: Solid lithium exists in a metallic lattice. But how do these atoms stay together? Let's look closer at lithium's electrons.

In the next clip, a single Li atom is depicted. On-screen text reads: Electron configuration: $1s^2 2s^1$. $1s^2$ are the core electrons. $2s^1$ is the valence electron. When metal atoms are arranged in a lattice, the outer valence electrons are not fixed to an individual lithium atom. Instead, these valence electrons are shared with all the atoms in the metallic lattice.

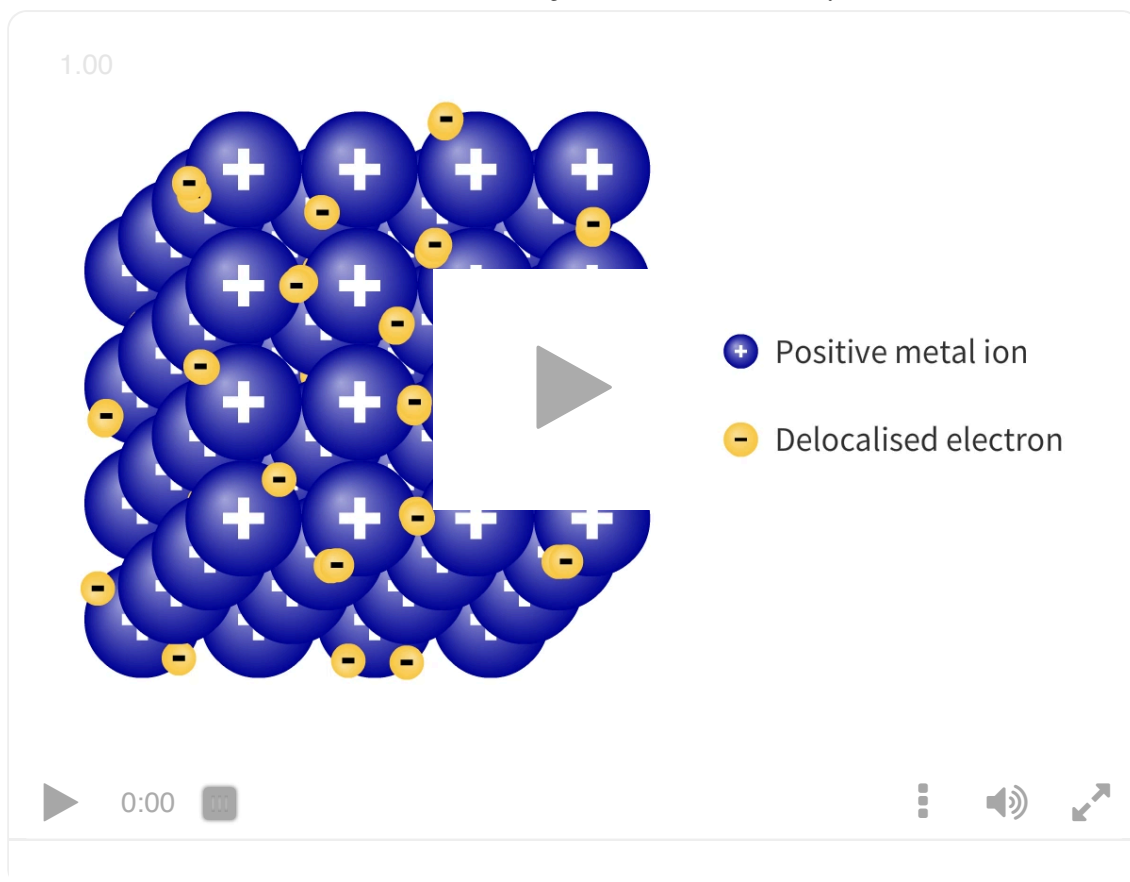
Next clip: A lithium ion with a positive charge and a valence electron outside it. The valence electron is indicated using an arrow. The text reads as follows: This makes the metal atoms in the lattice positively charged. The valence electron is free to move throughout the lattice, meaning it is delocalised.

Next clip: A lattice with four rows and four columns of Li^+ ions, each with a valence electron around it. The Li^+ ions are represented as circles, and the electrons are represented as dots. The text reads as follows: The metallic lattice is made up of metal cations arranged in a regular pattern with delocalised valence electrons freely moving between them.

Next clip: Four Li^+ ions with their valence electrons. Double-headed arrows are placed between the ions and nearby valence electrons. The text on the left reads as follows: There is a strong electrostatic attraction between positive metal ions and delocalised valence electrons. This electrostatic attraction is how metal atoms stay together in a lattice. This is the metallic bond.

The interactive helps users understand the formation of metallic bond in lithium atoms.

When metal atoms arrange in a lattice, the outer valence electrons are strongly attracted to neighbouring atoms, moving through the empty spaces in the lattice away from the atom it was originally part of. These electrons are shared among all metal atoms in the lattice. As the valence electrons are not bound to any one specific atom they are called delocalised electrons and free to move throughout the lattice. As valence electrons are delocalised, the remaining metal atoms in the lattice become positively charged cations. These cations repel each other, because they have the same charge, and keep the neatly arranged lattice in place. There are very strong electrostatic forces between the lattice of positive metal cations and the delocalised electrons: this is the metallic bond and is shown in **Interactive 3**.



Interactive 3. Structure of Metals.

 More information for interactive 3

An interactive video exhibits a cubic metal lattice structure with positive metal ions arranged in an orderly manner. Positive ions are indicated as purple circles with a plus sign in the middle. Delocalized electrons move randomly through the empty spaces within the lattice. These electrons are indicated as yellow circles with a minus sign in the middle. Delocalized electrons, also known as valence electrons, are shared among all the metal ions in the lattice.

The “symbols” to identify the positive and negative ions in the lattice are provided on the right side of the lattice structure.

The play button with a slider allows the users to play and pause the video as required. In the bottom right, the speedometer, sound, setting, and zoom-out icons are provided consecutively. Except for the setting icon, all other icons are highlighted in white. This video helps users understand the structure of the metal and the movement of delocalized electrons.

In **Interactive 4**, drag and drop the box with the electron configuration, number of valence electrons and the structure into the correct location in the table.

Metal	Sodium	Aluminium	Magnesium
Electron configuration			
# valence electrons			
Structure			
	$1s^2 2s^2 2p^6 3s^2$	$1s^2 2s^2 2p^6 3s^1$	$1s^2 2s^2 2p^6 3s^2 3p^1$
	1	2	3
☒ Check			

Interactive 4. Electron Configurations and Valence Electrons for Different Structures.

More information for interactive 4

An interactive table features labels on the top and the left with missing fields. The labels on the top from left to right read: Metal, Sodium, Aluminum, and Magnesium. The labels on the left from top to bottom read: Electron configuration, # valence electrons, and Structure. There are three missing rows and columns in the interactive table.

Below the table, there are three rows and three columns with draggable options. The options in the first row from left to right include: $1s^2 2s^2 2p^6 3s^2$, $1s^2 2s^2 2p^6 3s^1$, and

$1s^2 2s^2 2p^6 3s^2 3p^1$. The options in the second row from left to right include: 1, 2, and 3.

The options in the third row from left to right include:

A square box with orange circles (metal ions) labeled with $3+$ that are surrounded by a sea of electrons in peacock green.

A square box with pink circles (metal ions) labeled with $+$ that are surrounded by a sea of electrons in peacock green.

A square box with blue circles (metal ions) labeled with $2+$ that are surrounded by a sea of electrons in peacock green.

This interactivity helps users understand metallic structures, valence electrons, and electron configurations for sodium, aluminum, and magnesium. The “Check” icon at the bottom left allows users to check their answers. The right answer is indicated with a slider. If the inputted answers are wrong, the “Retry” icon at the bottom left allows users to retry the activity.

Read below for the solution.

Sodium

Electronic configuration: $1s^2 2s^2 2p^6 3s^1$

Number of valence electrons: 1

Structure: A square box with pink circles (metal ions) labeled with $+$ that are surrounded by a sea of electrons in peacock green.

Aluminum

Electronic configuration: $1s^2 2s^2 2p^6 3s^2 3p^1$

Number of valence electrons: 3

Structure: A square box with orange circles (metal ions) labeled with $3+$ that are surrounded by a sea of electrons in peacock green

Magnesium

Electronic configuration: $1s^2 2s^2 2p^6 3s^2$

Number of valence electrons: 2

Structure: A square box with blue circles (metal ions) labeled with $2+$ that are surrounded by a sea of electrons in peacock green.



Exercise 1

Click a question to answer



1. True or false?



Metallic bonding is the electrostatic attraction between cations in the three-dimensional lattice.



Check answers

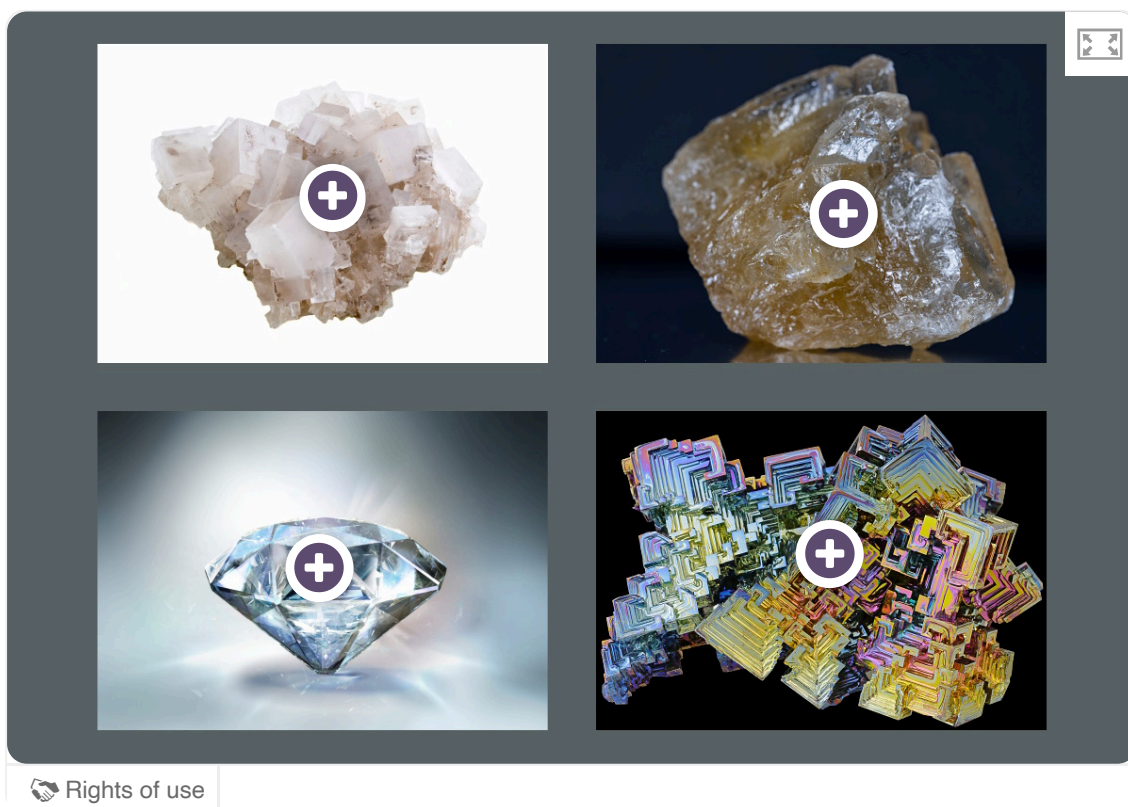
Nature of Science

Aspect: Models

This metallic structure and bonding model is known as the 'free electron model'. While it is incredibly useful to help understand the physical properties of metals through considering ions at the atomic scale, this model also has its limitations. For example, the repulsive interaction between the valence electrons is mostly neglected. One way to identify unbiased information is the acknowledgement of limitations.

Metallic bonding is unique from its ionic and covalent counterparts as the bonding is not between atoms/ions. Instead, the bonding occurs between the cations within the lattice structure and the delocalised valence electrons.

Interactive 5 shows a comparison of the structures for all three types of bonding. These valence electrons act similarly to a glue keeping the cations bonded together. This type of bonding is non-directional because the electrostatic attractive force occurs in all directions.



Interactive 5. Crystalline Solids of Sodium Chloride, Sugar, Diamond and Bismuth.

 More information for interactive 5

An interactive collage displays four photographs of crystalline solids. Each photograph has a plus sign representing an interactive hotspot. Selecting a hotspot reveals the type of bonding and the internal structure of the corresponding material.

Top left:

Photo: A chunk of rock salt displaying intergrown white to transparent crystals.

Hotspot: Ionic. A cubic structure of sodium chloride featuring an alternating arrangement of sodium and chloride ions. Chloride ions are represented by large spheres, and sodium ions are represented by small spheres.

Top right:

Photo: An irregularly shaped golden brown sugar crystal.

Hotspot: Covalent-small molecule. The molecular structure of sugar depicts repeating sucrose units with carbon, hydrogen, and oxygen atoms.

Bottom left:

Photo: A single crystal of diamond displaying its shiny, reflective surface.

Hotspot: Covalent network. A cubic structure illustrates carbon atoms occupying the corners and face centers of the cube. Each carbon atom is bonded to four other carbon atoms in a tetrahedral geometry.

Bottom right:

Photo: A bismuth crystal exhibits step-like structures and iridescent colors. The crystal is shiny and has sharp edges.

Hotspot: Metallic: A cubic structure of arranged positive metal ions surrounded by a sea of delocalized negative electrons.

This interactive collage helps users identify the types of bonding in crystalline solids and their internal arrangements.

It is important to note that although the structure of metals involves metal cations, metals themselves are electrically neutral because all valence electrons remain part of the overall structure.



Exercise 2

Click a question to answer



1. Which of the following species are involved in metallic bonding:



- I. Cations
- II. Core electrons
- III. Valence electrons

[✔ Check answers](#)

Activity

- **IB learner profile attribute:** Communicator
- **Approaches to learning:**
 - Thinking skills — Applying key ideas and facts in new contexts
 - Communication skills — Clearly communicating complex ideas in response to open-ended questions
- **Time required to complete activity:** 20—30 minutes
- **Activity type:** Pair activity

Balloon bonding models. Each student will need one balloon.

In pairs, use your balloons to model the three types of bonding. Your balloon will represent a valence electron and your body represents the nucleus of an atom. The combination of you and your balloon is neutral.

- How might you use the balloons to demonstrate ionic bonding with your partner? What process needs to occur with neutral atoms before an ionic compound can be formed?
- How might you use the balloons to demonstrate covalent bonding with your partner? What occurs between two atoms when a covalent bond is formed?
- How might you use the balloons to demonstrate metallic bonding? What are valence electrons doing in a metal?

Discuss the following with your partner:

- In each type of bonding, identify and explain the source of attraction between particles. How did you show this with your balloon model?

Compare and contrast metallic bonding with the other types of bonding using a Venn diagram:

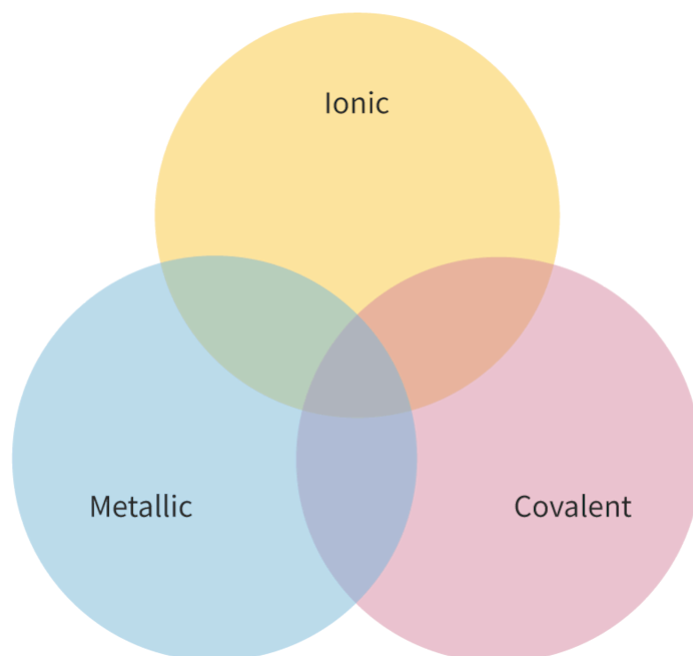


Figure 2. Venn diagram for comparing and contrasting metallic bonding with the other types of bonding.

 More information for figure 2

The image is a Venn diagram that compares and contrasts three types of chemical bonding: metallic, ionic, and covalent. The diagram consists of three intersecting circles. Each circle represents one type of bonding. The first circle, labeled 'Metallic,' overlaps with the other two circles. The second circle, labeled 'Ionic,' overlaps with the 'Metallic' and 'Covalent' circles. The third circle is labeled 'Covalent.' The overlaps in the Venn diagram represent shared characteristics between the bonding types. For example, where the 'Metallic' and 'Ionic' circles overlap, common properties shared by metallic and ionic bonding are indicated. The central area, where all three circles intersect, would show characteristics that all three types of bonding share. However, the specific shared characteristics are not listed in the diagram and require external knowledge or context to fill in.

[Generated by AI]

Extension

- **IB learner profile attribute:** Communicator
- **Approaches to learning:**
 - Thinking skills — Applying key ideas and facts in new contexts
 - Communication skills — Clearly communicating complex ideas in response to open-ended questions

- **Time required to complete activity:** 1 hour
- **Activity type:** Group/pair activity

Create a stop motion film to show the structure of a metallic lattice with delocalised electrons. Your target audience are primary school students. Use any material you have available to do so. There are some easy-to-find websites (e.g. [Flipgrid](https://my.flipgrid.com/)  (<https://my.flipgrid.com/>)) and social media tools that you can use to upload images and create a stop motion film.